$\boxed{1}$

(1)

$$
\begin{aligned}
\boldsymbol{A}\cdot(\boldsymbol{B}\times\boldsymbol{C}) &= \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \cdot \begin{pmatrix} B_yC_z - B_zC_y \\ B_zC_x - B_xC_z \\ B_xC_y - B_yC_x \end{pmatrix} \\
&= A_x\left(B_yC_x - B_zC_y\right) + A_y\left(B_zC_x - B_xC_z\right) + A_z\left(B_xC_y - B_yC_x\right) \\
&= A_xB_yC_z - A_xB_zC_y + A_yB_zC_x - A_yB_xC_z + A_zB_xC_y - A_zB_yC_x \qquad (1) \\
&= \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}
\end{aligned}
$$

行列式の展開式 (1) から

$\boldsymbol{B}$ の成分でまとめると

$$
\begin{aligned}
&B_x\left(C_yA_z - C_zA_y\right) + B_y\left(C_zA_x - C_xA_z\right) + B_z\left(C_xA_y - C_yA_x\right) \\
&= \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \cdot \begin{pmatrix} C_yA_z - C_zA_y \\ C_zA_x - C_xA_z \\ C_xA_y - C_yA_x \end{pmatrix} \\
&= \boldsymbol{B}\cdot(\boldsymbol{C}\times\boldsymbol{A})
\end{aligned}
$$

$\boldsymbol{C}$ の成分でまとめると

$$
\begin{aligned}
&C_x\left(A_yB_z - A_zB_y\right) + C_y\left(A_zB_x - A_xB_z\right) + C_z\left(A_xB_y - A_yB_x\right) \\
&= \begin{pmatrix} C_x \\ C_y \\ C_z \end{pmatrix} \cdot \begin{pmatrix} A_yB_z - A_zB_y \\ A_zB_x - A_xB_z \\ A_xB_y - A_yB_x \end{pmatrix} \\
&= \boldsymbol{C}\cdot(\boldsymbol{A}\times\boldsymbol{B})
\end{aligned}
$$

以上から、$\boldsymbol{A}\cdot(\boldsymbol{B}\times\boldsymbol{C}) = \boldsymbol{B}\cdot(\boldsymbol{C}\times\boldsymbol{A}) = \boldsymbol{C}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$

(2)

$$
\begin{aligned}
\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) &= \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \times \begin{pmatrix} B_y C_z - B_z C_y \\ B_z C_x - B_x C_z \\ B_x C_y - B_y C_x \end{pmatrix} \\
&= \begin{pmatrix} A_y \left(B_x C_y - B_y C_x\right) - A_z \left(B_z C_x - B_x C_z\right) \\ A_z \left(B_y C_z - B_z C_y\right) - A_x \left(B_x C_y - B_y C_x\right) \\ A_x \left(B_z C_x - B_x C_z\right) - A_y \left(B_y C_z - B_z C_y\right) \end{pmatrix} \\
&= \begin{pmatrix} A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z \\ A_z B_y C_z - A_z B_z C_y - A_x B_x C_y + A_x B_y C_x \\ A_x B_z C_x - A_x B_x C_z - A_y B_y C_z + A_y C_z B_y \end{pmatrix} \\
&= \begin{pmatrix} B_x \left(A_y C_y + A_z C_z\right) - C_x \left(A_y B_y + A_z B_z\right) \\ B_y \left(A_z C_z + A_x C_x\right) - C_y \left(A_z B_z + A_x B_x\right) \\ B_z \left(A_x C_x + A_y B_y\right) - C_z \left(A_x B_x + A_y B_y\right) \end{pmatrix} \\
&= \begin{pmatrix} B_x \left(A_x C_x + A_y C_y + A_z C_z\right) - A_x B_x C_x - C_x \left(A_x C_x + A_y C_y + A_z C_z\right) + A_x B_x C_x \\ B_y \left(A_x C_x + A_y C_y + A_z C_z\right) - A_y B_y C_y - C_y \left(A_x C_x + A_y C_y + A_z C_z\right) + A_y B_y C_y \\ B_z \left(A_x C_x + A_y C_y + A_z C_z\right) - A_z B_z C_z - C_z \left(A_x C_x + A_y C_y + A_z C_z\right) + A_z B_z C_z \end{pmatrix} \\
&= \begin{pmatrix} B_x \left(A_x C_x + A_y C_y + A_z C_z\right) - C_x \left(A_x C_x + A_y C_y + A_z C_z\right) \\ B_y \left(A_x C_x + A_y C_y + A_z C_z\right) - C_y \left(A_x C_x + A_y C_y + A_z C_z\right) \\ B_z \left(A_x C_x + A_y C_y + A_z C_z\right) - C_z \left(A_x C_x + A_y C_y + A_z C_z\right) \end{pmatrix} \\
&= \left(A_x C_x + A_y C_y + A_z C_z\right) \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} - \left(A_x B_x + A_y B_y + A_z B_z\right) \begin{pmatrix} C_x \\ C_y \\ C_z \end{pmatrix} \\
&= \left(\boldsymbol{A} \cdot \boldsymbol{C}\right) \boldsymbol{B} - \left(\boldsymbol{A} \cdot \boldsymbol{B}\right) \boldsymbol{C}
\end{aligned}
$$

$\boxed{2}$

(1)

余弦定理より、$\left|\boldsymbol{r}-\boldsymbol{r}'\right|^2 = r^2 - 2\left(\boldsymbol{r}\cdot\boldsymbol{r}'\right) + r'^2$ が成り立つ。

$$\begin{aligned}
\frac{1}{\left|\boldsymbol{r}-\boldsymbol{r}'\right|^3} &= \left|\boldsymbol{r}-\boldsymbol{r}'\right|^{-3} \\
&= \left(r^2 - 2\left(\boldsymbol{r}\cdot\boldsymbol{r}'\right) + r'^2\right)^{-\frac{3}{2}} \\
&= r^{-3}\left(1 - 2\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} + \frac{r'^2}{r^2}\right)^{-\frac{3}{2}}
\end{aligned}$$

ここで、$r \gg r'$ より、$\left|\dfrac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2}\right| = \left|\dfrac{rr'\cos\theta}{r^2}\right| = \left|\dfrac{r'}{r}\cos\theta\right| \ll 1$ である。r' の一次近似により r' の 2 次スケールを持つ第 3 項は無視される。

$$\approx \frac{1}{r^3}\left(1 + 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2}\right)$$

(2)

(1) で求めた近似式を用いて積分式を展開する。

$$\begin{aligned}
\boldsymbol{E}\left(\boldsymbol{r}\right) &= \frac{1}{4\pi\varepsilon_0}\int_\Omega \frac{\boldsymbol{r}-\boldsymbol{r}'}{\left|\boldsymbol{r}-\boldsymbol{r}'\right|^3}\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' \\
&\approx \frac{1}{4\pi\varepsilon_0}\int_\Omega \frac{1}{r^3}\left(1 + 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2}\right)\left(\boldsymbol{r}-\boldsymbol{r}'\right)\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' \\
&= \frac{1}{4\pi\varepsilon_0}\int_\Omega \left(\frac{1}{r^3} + 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\right)\left(\boldsymbol{r}-\boldsymbol{r}'\right)\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' \\
&= \frac{1}{4\pi\varepsilon_0}\int_\Omega \left(\frac{\boldsymbol{r}}{r^3} - \frac{\boldsymbol{r}'}{r^3} + 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\boldsymbol{r} - 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\boldsymbol{r}'\right)\rho\left(\boldsymbol{r}'\right)\mathrm{d}V'
\end{aligned}$$

ここでも、r' の 2 次スケールを持つ第 4 項 $-3\dfrac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\boldsymbol{r}'$ は無視される。

$$\begin{aligned}
&\approx \frac{1}{4\pi\varepsilon_0}\int_\Omega \left(\frac{\boldsymbol{r}}{r^3} - \frac{\boldsymbol{r}'}{r^3} + 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\boldsymbol{r}\right)\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' \\
&= \frac{1}{4\pi\varepsilon_0}\int_\Omega \frac{\boldsymbol{r}}{r^3}\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' - \frac{1}{4\pi\varepsilon_0}\int_\Omega \frac{\boldsymbol{r}'}{r^3}\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' + \frac{1}{4\pi\varepsilon_0}\int_\Omega 3\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^5}\boldsymbol{r}\rho\left(\boldsymbol{r}'\right)\mathrm{d}V' \\
&= \frac{1}{4\pi\varepsilon_0}\frac{\boldsymbol{r}}{r^3}\int_\Omega \rho\left(\boldsymbol{r}'\right)\mathrm{d}V' - \frac{1}{4\pi\varepsilon_0}\frac{1}{r^3}\int_\Omega \rho\left(\boldsymbol{r}'\right)\boldsymbol{r}'\,\mathrm{d}V' + \frac{3}{4\pi\varepsilon_0}\frac{1}{r^5}\boldsymbol{r}\cdot\int_\Omega \rho\left(\boldsymbol{r}'\right)\boldsymbol{r}'\,\mathrm{d}V'\boldsymbol{r} \\
&= \frac{Q}{4\pi\varepsilon_0}\frac{\boldsymbol{r}}{r^3} - \frac{\boldsymbol{p}}{4\pi\varepsilon_0 r^3} + \frac{3\left(\boldsymbol{r}\cdot\boldsymbol{p}\right)}{4\pi\varepsilon_0 r^5}\boldsymbol{r}
\end{aligned}$$